Junseo Lee

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Education

B.S. in Naval Architecture and Ocean Engineering Mar. 2022 – Feb. 2028 (Expected)
Seoul National University (SNU) Seoul, Korea

- GPA: 3.35/4.30 (Major: 3.44/4.30)
- Relevant coursework: Structural Mechanics, Structural Vibrations, Finite Element Theory, and Ship Design

Research Experience

Undergraduate Intern Aug. 2026 – Present
OST Lab., Seoul National University • Advisor: Prof. Dokyun Kim Seoul, Korea

- AI-based double hull tanker midship optimisation tool development
- Multi-fidelity fluid–structure interaction analysis

Undergraduate Visiting Researcher Aug. 2025 – Feb. 2026
MORIN Lab., KAIST • Advisor: Prof. Jinwhan Kim Daejeon, Korea

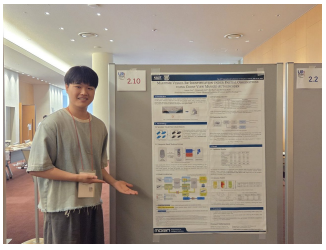
- Organized existing maritime sensor datasets for model development and evaluation, and participated in field experiments at Jeongok Port.
- **LiDAR-based maritime vessel Re-Identification (ReID):** Developed CIEF-Net, a neural network that integrates local curvature features with spatial coordinates through early fusion for vessel re-identification.
- **Cross-View Representation Learning under Partial Observations:** Developed a cross-view masked autoencoder to learn vessel representations by reconstructing a target view from partial observations of another viewpoint.

Publications & Presentations

Development of an AI-Based Midship Section Optimal Design Support Program Oct. 2026
***Junseo Lee**, Woojin Jeong, and Seungwoo Jin*
Poster Presentation, CAE Contest Poster Session • *KSOE 2026 Fall Conference*



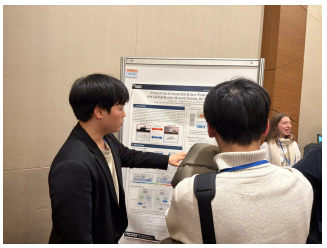
Maritime Vessel Re-Identification under Partial Observations Using Cross-View Masked Autoencoder Jul. 2026
***Junseo Lee**, Dongwook Lee, Sol Han, and Jinwhan Kim* Osaka, Japan
Poster Presentation • *UR 2026*



Curvature-Integrated Early Fusion Network for LiDAR-Based Marine Vessel Re-Identification Feb. 2026

Junseo Lee, Dongwook Lee, and Jinwhan Kim PyeongChang, Korea

Poster Presentation • *KROS 2026*



Awards & Honors

ABS Scholarship Oct. 2026

Funding

2nd Prize, Ocean Engineering CAE Contest (KSOE) Sep. 2026

Team Leader, LLM-based midship optimisation software development



JeongGong Scholarship Jun. 2026

Funding

DSME Prize, Autonomous Ship Contest for Freshmen (SNU) Jun. 2022

Developing collision detection algorithm

Skills

Programming	Python, MATLAB, C/C++
Machine Learning	PyTorch, Open3D
Languages	Korean (Native), English (Fluent)

Military Service

DMZ East Corridor Patrol Feb. 2024 – Aug. 2025

Sergeant (discharged)

- Commendation for Successful Defection Inducement Operation (CJCS, Aug. 2024)
- Certificate of Achievement (UNCMAC-S, Aug. 2025)

Research Interests

My research interests lie at the intersection of machine learning and naval architecture, with a focus on integrating physical principles and domain knowledge to support reliable analysis, interpretable predictions and informed design decisions. In maritime perception, I study self-supervised learning of 3D geometric representations that remain robust to partial observations, viewpoint changes and limited real-world data. In structural design, I am interested in multi-fidelity surrogate modelling with active learning to reduce the cost of fluid–structure interaction analysis, and in optimisation that embeds classification rules such as the IACS Common Structural Rules directly into the design process. Across both areas, I seek methods whose limitations are well understood and whose results can be checked against physical reasoning and engineering evidence.